



Project Title	Rising Waters: Machine Learning Based Flood Prediction System		
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7. Technology Stack

The stack was chosen for minimal operational overhead: every layer below is either part of the Python data-science standard toolkit or a dependency-free frontend technology, which keeps the project runnable with a single pip install and python app.py, with no databases, message queues, or external services to configure.

Layer	Technology
Backend / Web Framework	Python 3, Flask
Machine Learning	Scikit-learn (Decision Tree, Random Forest, KNN), XGBoost
Data Handling	Pandas, NumPy
Model Serialization	Joblib (flood_model.joblib, scaler.joblib)
Frontend	HTML5, Vanilla CSS3 (glassmorphism styling), JavaScript
Exploratory Analysis	Jupyter Notebook (notebook.ipynb)
Dataset	flood_prediction.csv — historical rainfall/weather records with binary flood label
Tooling / Version Control	VS Code, GitHub

Scikit-learn and XGBoost together cover the full comparison set used in train.py, while Pandas and NumPy handle all tabular manipulation both offline (training) and online (assembling the single-row prediction Data Frame in app.py). Joblib was chosen over alternatives such as pickle specifically because it is the format scikit-learn's own documentation recommends for persisting fitted estimators, and it handles NumPy arrays more efficiently.

